Topic detail

Yes. For your research topic **“Illumination-Aware Evaluation of Object Detection in Low-Light Conditions Using the ExDark Dataset”**, there is a good set of recent literature from **2024 onward**, especially papers that explicitly use **ExDark**.

Most relevant papers to cite

Year	Paper	Venue	Why it is useful for your topic
2026	**Su, Y., & Lu, M. – “Exposure-Aware Training for Low-Light Object Detection Without Target-Domain Data”**	*Journal of Imaging*	Very relevant to **illumination/exposure-aware evaluation**. Uses illumination attenuation and noise degradation and evaluates on ExDark.
2026	**Singh, S., Kumari, R., Pallavi, P., et al. – “A systematic review of deep learning methods for low-light image enhancement and object detection”**	*Discover Applied Sciences*	Excellent recent **literature review**. It experimentally evaluates **YOLOv8–YOLOv11 on ExDark** and discusses metrics, datasets, challenges, and future research.
2026	**Sharif, S. M. A., Rehman, A., Abidin, Z. U., et al. – “Illuminating Darkness: Learning to Enhance Low-light Images In-the-Wild”**	**WACV 2026**	Important for the **illumination-enhancement → detection** relationship; reports a **+6.80 mAP improvement on ExDark** after enhancement.
2025	**Ye, S., Huang, W., Liu, W., Chen, L., Wang, X., & Zhong, X. – “YES: You should Examine Suspect cues for low-light object detection”**	*Computer Vision and Image Understanding*	Probably one of the **best matches** for “illumination-aware.” It explicitly addresses **uneven lighting**, object-background confusion, and uses ExDark.
2025	**Ren, B., Xu, Z., Zhao, J., Hao, R., & Zhang, J. – “Complex dark environment-oriented object detection method based on YOLO-AS”**	*Scientific Reports*	Uses **ExDark and LOL**, evaluates with **mAP@50 and mAP@50:95**, and focuses on complex/dark illumination conditions.
2025	**“YOLO-D: A Domain Adaptive approach**	*Procedia Computer*	Uses ExDark and specifically addresses **different illumination**

Year	Paper	Venue	Why it is useful for your topic
	towards low light object detection"	*Science*	**levels / domain shifts** using a low-light enhancement and domain-adaptation framework.
2024	**Peng, D., Ding, W., & Zhen, T. – "A novel low light object detection method based on the YOLOv5 fusion feature enhancement"**	*Scientific Reports*	Directly uses **ExDark** and proposes noise suppression, feature enhancement, attention and a decoupled detection head. Very useful for your methodology/background.
2024	**Li, Z., Xiang, J., & Duan, J. – "A low illumination target detection method based on a dynamic gradient gain allocation strategy"**	*Scientific Reports*	Extremely relevant: explicitly studies **low illumination**, uses ExDark, and introduces a dynamic strategy for low-light localization. Reports **75.60% mAP@50** on ExDark.
2024	**"RFSC-Net: Re-parameterization forward semantic compensation network in low-light environments"**	*Image and Vision Computing*	Uses **ExDark** and focuses on retaining detailed/semantic information under low illumination; reports improvement over YOLOv8n.
2024	**"Advanced Object Detection in Low-Light Conditions: Enhancements to YOLOv7 Framework"**	*Remote Sensing*	Direct ExDark study with brightness augmentation, attention, deformable convolution and a modified loss. Reports **mAP50 = 80.1%** and **mAP50:95 = 52.3%**.
2024	**"Object detection in low light environments"**	*Procedia Computer Science*	Useful as a straightforward benchmark/reference study using **ExDark + YOLOv8**, preprocessing and transfer learning.
2024	**Du, Z., Shi, M., & Deng, J. – "Boosting Object Detection with Zero-Shot Day-Night Domain Adaptation"**	**CVPR 2024**	High-quality conference reference. Uses **ExDark** and studies **day-to-night/illumination domain adaptation** using Retinex-based illumination invariance.

The strongest papers for your exact title

For a thesis/project specifically called **"Illumination-Aware Evaluation of Object Detection in Low-Light Conditions Using the ExDark Dataset,"** I would prioritize these six:

1. Ye et al. (2025) – YES
Best conceptual connection to **uneven illumination** and the interaction between object and

background.

2. Li et al. (2024) – DimNet / dynamic gradient gain allocation
Very useful for discussing how **illumination affects localization and detection accuracy**.

3. Du et al. (2024) – Zero-Shot Day-Night Domain Adaptation
Excellent reference if your “illumination-aware” idea includes **day/night or illumination-domain variation**. It is also a strong CVPR reference.

4. Su & Lu (2026) – Exposure-Aware Training
Particularly useful if you plan to divide ExDark images by **exposure/illumination severity** and compare detector robustness.

5. Singh et al. (2026) – Systematic review
Very useful for your **literature review chapter**, research gap, datasets, metrics, and identification of recent methods.

6. Peng et al. (2024) – NLE-YOLO
A good baseline/reference for an ExDark-based experimental setup and low-light-specific detector architecture.

Useful links / DOI information

Peng et al., 2024 — *A novel low light object detection method based on the YOLOv5 fusion feature enhancement*
DOI: **10.1038/s41598-024-54428-8**

Li et al., 2024 — *A low illumination target detection method based on a dynamic gradient gain allocation strategy*
DOI: **10.1038/s41598-024-80265-w**

Du et al., 2024 — *Boosting Object Detection with Zero-Shot Day-Night Domain Adaptation*
DOI: **10.1109/CVPR52733.2024.01204**

Ye et al., 2025 — *YES: You should Examine Suspect cues for low-light object detection*
DOI: **10.1016/j.cviu.2024.104271**

Ren et al., 2025 — *Complex dark environment-oriented object detection method based on YOLO-AS*
DOI: **10.1038/s41598-025-07348-0**

Su & Lu, 2026 — *Exposure-Aware Training for Low-Light Object Detection Without Target-Domain Data*
DOI: **10.3390/jimaging12060245**

A useful research gap for your proposed study

A promising distinction for your work is that many recent papers **propose a new detector or enhancement method and report overall mAP**, but fewer studies make **illumination severity itself the central evaluation variable**. For example, recent work explicitly discusses uneven lighting, exposure degradation, or day/night domain shifts, while ExDark contains images covering a range of low-light conditions.

That gives you a strong angle: evaluate **how detection performance changes as illumination severity changes**, rather than only reporting one aggregate ExDark score. You could compare **mAP@0.5, mAP@0.5:0.95, precision, recall, F1, and miss rate** across illumination categories/severity levels. The 2026 systematic review is especially useful here because it discusses evaluation metrics and reports experiments across ExDark.

I can also turn these into a **20-paper literature review table with APA/IEEE citations, DOI links, dataset, model, metrics, results, and research gap**, ready to paste into your thesis.